Complaint Compass: market opportunity and scalability strategy

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Executive summary

Complaint Compass targets a $2.6–2.9B complaint-management software market (S5, S6). It enters through Indian quick commerce, where order volume is high and statutory clocks are short. It scales by adding workers, not by redesigning. Every figure below is tagged with a source, and each source is quoted in the Evidence section at the end.

1. **The need is growing and regulated.** Calls to India's National Consumer Helpline rose from 12,553 in December 2015 to 1,55,138 in December 2024 (S1). E-commerce sellers must acknowledge complaints within 48 hours (S4).
2. **The beachhead is concentrated and doubling.** Quick-commerce orders reached ₹64,000 crore in FY25, up from ₹30,000 crore (S9). Three platforms hold 98% of the market (S11).
3. **Our edge is measured.** Median filing takes 0.55 s, 96.1% of complaints never reach an LLM, and model cost is $4.33 per 100,000 complaints (R1, R2).
4. **Scale is a known path.** We run stateless intake workers in parallel and one writer per business × area. 10M complaints a month cost about $433 in model calls (R3).
5. **The open question is real-world accuracy.** Our test data is synthetic. We recommend a two-week shadow pilot with one operator before quoting accuracy or throughput as fact.

1. The problem: complaint volume is rising, and the law sets the clock

Complaints to Indian consumer businesses are rising across both consumer and regulated sectors. Statutory deadlines turn slow triage into a compliance risk.

Evidence	Figure	Source
Calls to the National Consumer Helpline	12,553 (Dec 2015) to 1,55,138 (Dec 2024), more than tenfold	S1 PIB, 14 Mar 2025
E-commerce grievances to the Ministry of Consumer Affairs	2,05,393 (FY21) to 4,45,960 (FY24)	S2 Storyboard18, citing the Ministry

Evidence	Figure	Source
Complaints under the RBI Integrated Ombudsman Scheme	13.34 lakh in FY2024-25, up 13.55%	S3 The Tribune (ANI), 1 Dec 2025
Deadline under the E-Commerce Rules, 2020	Acknowledge within 48 hours, resolve within one month	S4 IndiaLaw summary

Why it is hard to fix. Our own build shows the mechanics (R4):

- The same fault arrives in many wordings, across email, calls, chat and social, in English, Hindi and Hinglish.
- One customer often complains on several channels, so message counts overstate impact.
- New faults, such as a bad app release, appear before anyone has named them.

Sending every message to an LLM solves the reading problem. It does not solve speed, cost at volume, or dependence on one provider's uptime.

2. Market size: a $2.6–2.9B core market growing about 9–10% a year

Two independent analysts put complaint management software at $2.6–2.9B in 2025, growing 8.6–10.5% a year. AI-led segments grow roughly twice as fast. Each row below is one analyst's estimate, as stated on the publisher's own page.

Market	Analyst	2025 size	Forecast	CAGR	Source
Complaint management software	IMARC Group	$2.9B	$6.3B by 2034	8.57% (2026–34)	S5
Complaint management software	360iResearch	$2.63B ($2.89B in 2026)	$5.29B by 2032	10.49%	S6
AI in call centres	360iResearch	$1.70B	$5.20B by 2032	17.31%	S8
AI complaint management	IMARC Group	$6.41B	$33.19B by 2034	19.45% (2026–34)	S7

Reading the range. Our core range, $2.6–2.9B, comes from the first two rows. IMARC's AI complaint management figure ($6.41B) is larger than IMARC's own figure for the whole category ($2.9B). The two reports evidently use different scopes, so we do not add it to our estimate. AI in call centres is adjacent because voice is one of our four channels.

Estimates we could not verify and left out. Coherent Market Insights ($2.84B, 2025) and the wider customer-experience-management estimates (MarketsandMarkets, KBV Research, Fortune Business Insights) appeared in search results. Their pages refused automated access, so they are excluded.

3. Beachhead: Indian quick commerce first, then regulated fintech

Quick commerce is the best entry point: it is growing fast, highly concentrated, and its complaints are short, repetitive and time-critical.

Evidence	Figure	Source
Gross order value	₹30,000 crore (FY24) to ₹64,000 crore (FY25)	S9 PTI via Outlook Business, 10 Jul 2025
Projected gross order value	About ₹2 lakh crore by FY28 (CareEdge Ratings)	S9
Gross merchandise value	$1.6B (2022) to $7B (2024); over 40% a year expected to 2030	S10 Flipkart and Bain & Company
Daily orders	7.8 million a day (Jan 2026), about 95% year-on-year growth	S11 GAEE India, citing RedSeer
Market share, 2025	Blinkit 52%, Swiggy Instamart 25%, Zepto 21%, others 2%	S11

Illustrative sizing of complaint volume. This is our own calculation. The complaint rate is an assumption, so we show three rates rather than one. Orders are 7.8 million a day (S11) over 30 days. Costs use our measured $4.33 per 100,000 and the $12.32 per 100,000 cached-Haiku baseline (R2).

Share of orders that lead to a complaint (assumption)	1%	2%	3%
Complaints per month	2.3M	4.7M	7.0M
Our model cost per month	$101	$203	$304
LLM on every complaint, Claude Haiku with best caching	$288	$577	$865
Complaints avoided a month if fixing top issues removes 1%	23,400	46,800	70,200

Model cost is not the buyer's problem: contact volume is. The value of the last row is each avoided complaint times the operator's own cost per contact. We did not find a verifiable India figure for that cost, so we do not quote one.

Second wedge: fintech and banking. Of the 13.34 lakh RBI Ombudsman complaints in FY25, its central processing centre handled 9.11 lakh and Ombudsman offices received 2.96 lakh (S3). These are only the cases that escalated past the bank. Regulated firms need a documented trail per complaint, which our per-complaint record already provides (R4).

4. Competitive landscape: incumbents sell seats and suites; we sell triage per complaint

The major vendors price per agent seat or by custom enterprise quote. Our opening is a triage engine that sits underneath an existing helpdesk and is priced per complaint processed.

Vendor	Pricing evidence	What the source says it does	Source
Sprinklr	Self-serve sign-up retired; every suite now a custom Enterprise quote (the Service "Advanced" plan was about $249 a month before)	Unified CX suite: omnichannel inbox, knowledge base, chatbots	S12 eesel
Zendesk	Support Team from $19 per agent a month; Suite from $55 per agent a month	Ticketing across tickets, chat, email and messaging	S13 Chattermill
Zendesk QA	Per-agent add-on on top of a Zendesk Suite subscription	Quality assurance of agent conversations	S14 OnClarity
Medallia, Chattermill, Enterpret	Custom pricing based on feedback volume	Feedback analytics; Chattermill detects themes and sentiment automatically	S13
Qualtrics	Does not publish enterprise CX pricing	Survey-first experience management	S13

Vendor	Pricing evidence	What the source says it does	Source
Unwrap AI	Not stated	Clusters feedback by semantic meaning	S13

360iResearch's list of key companies in complaint management software includes Zendesk, Sprinklr, Salesforce, Medallia, Qualtrics, ServiceNow and Zoho (S6). Note that S12, S13 and S14 are blogs by vendors that compete with the products they describe.

Where we differ. Each point is backed by our own measured build (R1, R4). Public vendor material does not say whether incumbents do the same.

1. **Counts customers, not messages.** 100% of 1,215 customer links in the live run were correct, with 0 wrong. Ranking uses unique customers.
2. **The ranking is a published formula,** so every score can be checked by hand.
3. **Handles Hinglish.** Multilingual embeddings plus word matching gave 91% nearest-issue accuracy, against 80% and 84% for each alone.
4. **The LLM is the exception.** 96.1% of complaints never reached an LLM.
5. **Finds new problems without being told.** It found 5 of the 5 issues hidden in the test data.

Positioning: an engine that plugs into Zendesk, Freshdesk or an in-house CRM and feeds their dashboards, rather than a suite replacement.

5. Our edge: 3.5× faster at the median, with an LLM on only 3.9% of complaints

In a live run on 1,500 complaints with real models on 13 Nov 2026, the median complaint was filed in 0.55 s. An LLM call alone took 1.90 s on average. Every number below is in the committed run report (R1) or the session handoff (R2).

Measure (live run, 1,500 complaints)	Complaint Compass	LLM on every complaint	Source
Time to file, median	0.55 s	1.9 s or more	R2 §4, from per-call latencies
Complaints that never reach an LLM	96.1%	0%	R1 `share_without_llm` 0.9613

Measure (live run, 1,500 complaints)	Complaint Compass	LLM on every complaint	Source
Model cost, whole run	$0.065	Projected, not run	R2 §4
Model cost per 100,000	$4.33	$12.32 (Haiku, best caching) to $594 (Sonnet, no caching)	R2 §4
Filed under the right issue	98.4% (1,304 of 1,325)	Not run	R1 `accuracy.issue`
Real complaints kept / noise removed	99.9% / 100%	Not run	R1 `accuracy.gate`
Hidden problems found	5 of 5	Not run	R1 `accuracy.discovered`
First report to named issue, sudden faults	15.1 h (double charge), 23.4 h (app crash)	Depends on design	R2 §4

How we get there. Each complaint passes through checks ordered cheapest first. Shares come from the funnel in R1:

1. A decision model (Jev) drops non-complaints (10.1%), sorts the rest by business area and rates severity.
2. Free local similarity matching files 50.3%.
3. A Jev yes/no check against the top 3 candidates files 34.7%.
4. The LLM decides only 1.7%. Another 1.7% became new issues and 1.6% were still waiting.

What this table does not prove. The data is synthetic and template-based (R4 §12). The LLM-on-every-complaint baseline was priced from token counts calibrated on 12 real calls, not run (R2). Accuracy on real complaints will be lower.

6. Unit economics and pricing: $4.33 is our cost, not our price

The $4.33 per 100,000 complaints is what we pay model providers (Jev and Claude) to process complaints, measured on the live run. It is not a price. Our price is a proposal, set on the value to the customer and to be tested in the pilot.

What the $4.33 is made of. Cost per step, per 100,000 complaints, from the live run (R1 `cost.funnel`):

Step	Calls in the run	Cost per 100,000 complaints
Jev gate (every complaint)	1,500	$3.09
Jev verify	595	$0.64
LLM decisions	58	$0.49
Naming new issues, merging duplicates, group checks	19	$0.10
Total		**$4.33**

What it leaves out. Servers, the database, the CPU that runs the free similarity step, engineering and support. None of these was measured. In production, hosting will likely cost more than the model calls, so we do not quote a profit margin until Phase 1 measures it.

If real complaints are messier. Cost rises in a straight line, and slowly. One LLM decision cost $0.000128 on average (R1: $0.007418 over 58 calls). If 20% of complaints needed the LLM instead of 3.9%, the LLM line would rise from $0.49 to about $2.56. The total would go from $4.33 to about $6.40 per 100,000, still about half the cheapest LLM-on-everything option ($12.32, R2).

Proposed pricing: a platform fee plus a charge per complaint processed.

- **It contrasts with incumbents.** Zendesk charges $19–55 per agent per month (S13), and Sprinklr now sells only by custom quote (S12). Our price would grow with the customer's complaint volume, not their headcount.
- **It matches our costs,** which are per complaint.
- **The value is per complaint:** faster detection of new faults, and fewer repeat contacts once top issues are fixed.

Illustration (an assumption to test, not a decided price). This uses the 2% complaint rate from section 3.

Price per complaint (assumption)	$0.002	$0.005
Monthly revenue at 4.7M complaints	$9,400	$23,500
Our model cost at that volume	$203	$203

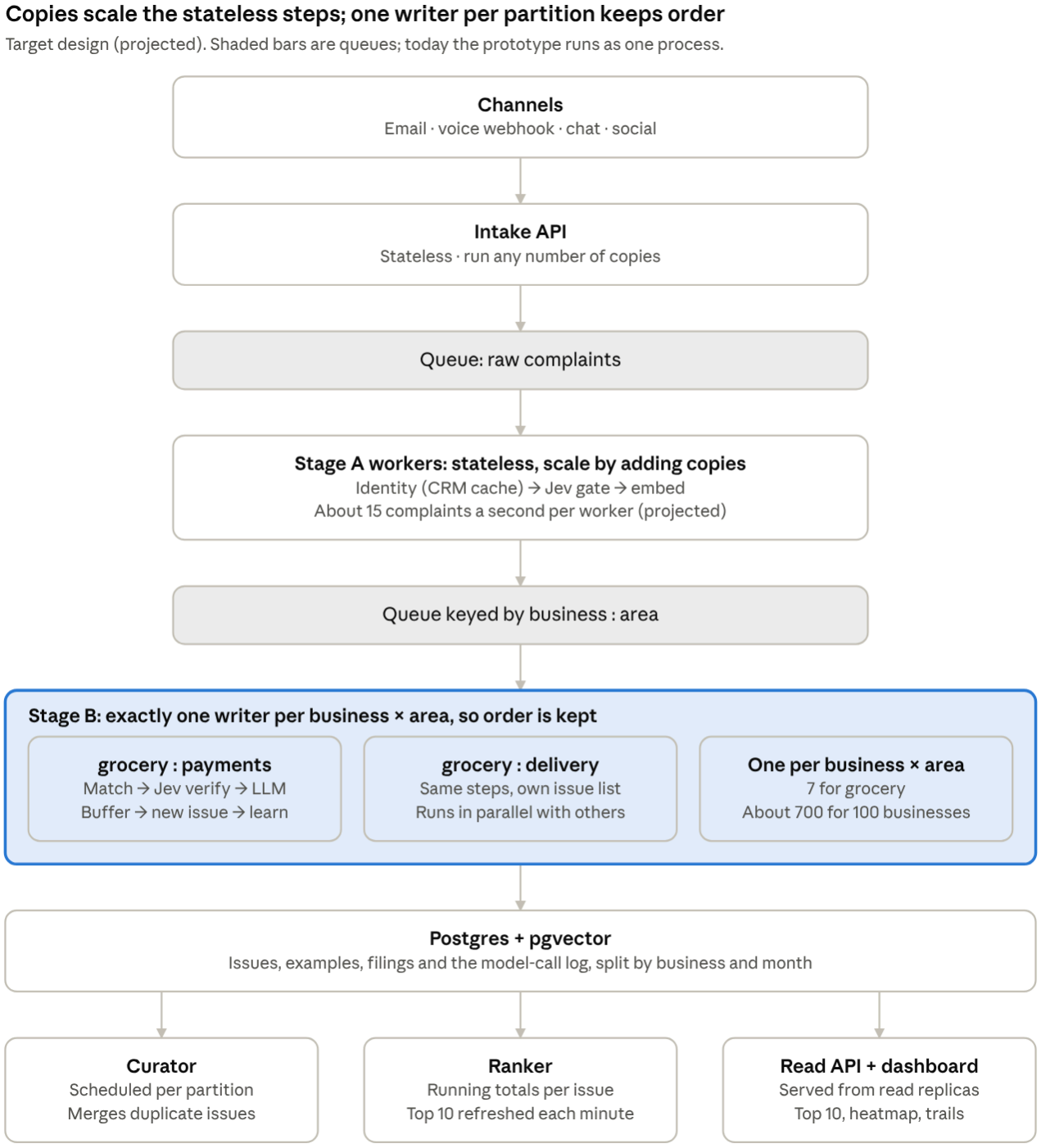
Price per complaint (assumption)	$0.002	$0.005
Model cost as a share of revenue	2.2%	0.9%

The real price point should come from the shadow pilot, measured against what the operator spends handling those complaints today.

One-line answer for judges: "Usage-based: a platform fee plus a charge per complaint. Our model cost is a fraction of a cent per complaint, so margin depends on hosting, which we measure in Phase 1."

7. Scalability strategy: split the funnel into stateless and partitioned stages

The design already scales in the ways that matter. 96.1% of complaints skip the LLM (R1), cost is linear in volume, and the only shared state, the issue list, is already grouped by business area in the engine (engine.py, `by_component`). Scaling means separating the steps that need no shared state from the one step that does. The full design is in R3.



Target architecture · stateless intake, partitioned single-writer funnel

Stage A depends only on the complaint itself, which is why today's engine can already run the gate ahead in 8 parallel threads (engine.py, `prefetch_gate`). We add copies until throughput is met. Stage B must handle each area's complaints in order, because the issue list grows as new problems are found. Keying the queue by business and area keeps that order within each partition while partitions run side by side. This is a design projection; it has not been built or load-tested.

8. What breaks first, and the fix

Each limit below is visible in the prototype's code, linked in the Evidence column. The "breaks at" figures are our projections from measured latencies; no load test has been run.

#	Limit in today's prototype	Evidence	Breaks at about	Fix
1	One global writer; issue list in one process's memory	api.py L38 `_lock`	A few complaints a second, sustained	One writer per business × area
2	SQLite, one file per run	db.py L43	Several writers, long history	Postgres with pgvector
3	Gate call 0.53 s on average, 8 in parallel	engine.py L143; latency in R2	About 15 a second, or 1.3M a day per worker (8 ÷ 0.53 s)	More workers; reuse answers for near-identical texts
4	Dense vectors of 16,768 numbers (384 + 16,384), scanned one by one	embed.py L68	1,000 issues × 25 examples × 16,768 × 4 bytes ≈ 1.7 GB in one area	Sparse word vectors; nearest-neighbour index
5	Waiting-tray scan grows with backlog	engine.py L208 `_to_buffer`	Large unmatched backlogs	30-day expiry; index on the tray
6	Top 10 recounted from all filings	ranker.py	Millions of filings	Running totals per issue, refreshed every minute
7	Jev is on the critical path	R4 §12	A Jev outage stops intake	Degraded mode: sort by similarity, re-check later
8	Provider rate limits	Not measured	Unknown	Queue back-pressure; second LLM provider

Throughput (projected from measured latencies).

- **Intake workers:** each handles about 15 complaints a second (8 parallel calls ÷ 0.53 s), so 10 workers handle about 13M a day.
- **Partition workers:** local matching takes about 0.02 s (R2), so one partition files about 50 complaints a second on the match path.

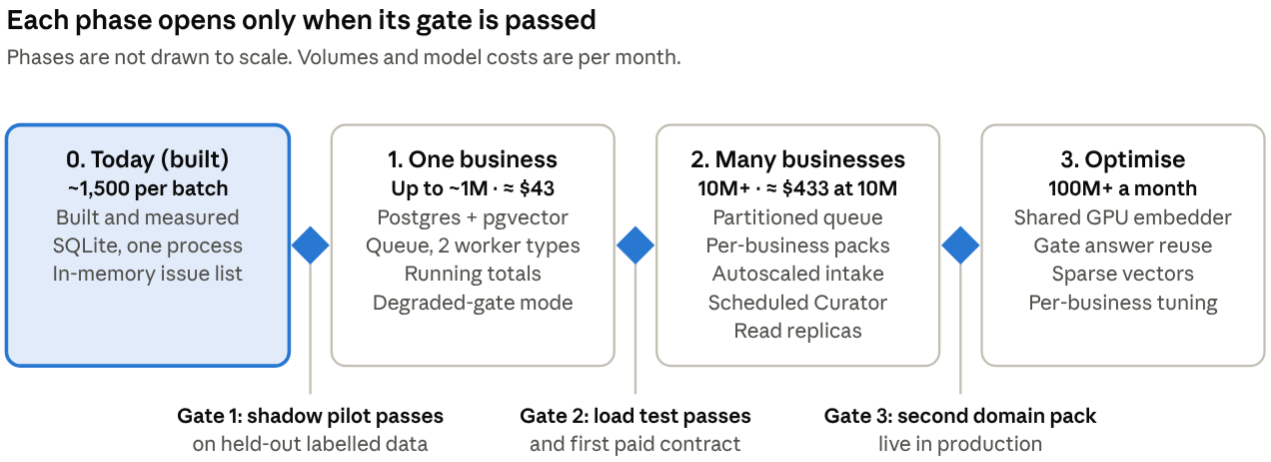
- **LLM traffic:** about 4% of volume, or 0.15 calls a second at 10M complaints a month. The queue absorbs spikes.

Model cost by volume. These figures are measured cost per complaint × volume, with hosting excluded (R2).

Complaints a month	Complaint Compass	Claude Haiku, best caching	Claude Sonnet, no caching
1M	$43	$123	$5,940
10M	$433	$1,232	$59,400
100M	$4,330	$12,320	$594,000

9. Roadmap: four phases, each unlocked by evidence

Each phase adds capacity only after the previous one proves itself on real data, so engineering spend stays behind demonstrated demand. Phase volumes and costs come from the cost-by-volume table in section 8.



Scaling roadmap · 4 phases, 3 gates

Phase 1 is the priority: it turns the prototype into a production service for one operator, and runs alongside the shadow pilot that clears Gate 1. The phases and gates are our plan, not a measured result.

10. Risks and recommendations

The largest risk is accuracy on real, messy complaints, not scale. Each risk below is either documented in our own build notes or follows from a gap in the evidence.

Risk	Evidence	Likelihood	Impact	Mitigation
Accuracy falls on real data	Data is template-generated (R4 §12.1)	High	High	Tune per customer on one labelled sample; measure on another
Cut-offs tuned on the data they were scored on	R4 §12.2	High	Medium	Report held-out accuracy only
Jev's endpoint is in alpha	R4 §12.6	Medium	High	Degraded mode; all gate logic in one file
Rate limits at volume are unknown	No load test run (section 8)	Medium	Medium	Load test before the first contract
Learned examples drift an issue	R4 §12.5; cap at engine.py L22	Low	Medium	Cap of 25 per issue; weekly re-check
Incumbents add similar triage	Large vendors already offer AI clustering and quality tools (S13, S14)	Medium	Medium	Integrate rather than compete; win on cost per complaint and auditability
Market figures differ by scope	Two analysts differ by $0.27B; a third segment exceeds its own parent (section 2)	Medium	Low	Quote ranges and name the analyst

Likelihood and impact ratings are our judgement.

Recommendations

1. **Lead with speed, support with cost.** Quote the measured 0.55 s and 96.1% figures, and name the comparison model every time.
2. **Run a shadow pilot.** Run alongside one quick-commerce operator for two weeks on real complaints, scored against a labelled sample.
3. **Price per complaint processed, not per seat.** This contrasts with the per-agent pricing of Zendesk and Sprinklr (S12, S13) and grows with the customer's volume.
4. **Build Phase 1 before the pilot ends:** Postgres, a queue, two worker types and degraded mode.

5. **Ship a second domain pack (fintech or telecom)** to prove the engine is domain-agnostic.
6. **Load-test before quoting throughput as fact.**

Evidence

Every external source below was opened on 13 Nov 2026. The quote is copied from the page as it read that day. Figures from analyst firms are from the publisher's public summary page, not the paid report.

ID	Source	Date	Quote from the page	Used in
S1	PIB, Ministry of Consumer Affairs: World Consumer Rights Day 2025	14 Mar 2025	"The number of calls received by NCH have grown more than tenfold, from 12,553 in December 2015 to 1,55,138 in December 2024."	Summary, 1
S2	Storyboard18: Flipkart, Amazon led e-com consumer complaints in FY24	FY24 data	"The Ministry of Consumer Affairs received a total of 4,45,960 grievances in the e-commerce sector alone in the fiscal year 2024 (FY24). This is a steep increase compared with 4,00,509 in FY23, 2,40,866 in FY22, and 2,05,393 grievances in FY21."	1
S3	The Tribune (ANI): RBI Ombudsman sees 13.34 lakh complaints in FY25	1 Dec 2025	"13.34 lakh complaints registered during FY 2024-25, a 13.55% increase from FY 2023-24"; "the Centralised Receipt and Processing Centre (CRPC) handled 9.11 lakh complaints, while Offices of the RBI Ombudsman (ORBIOs) received 2.96 lakh"	1, 3

ID	Source	Date	Quote from the page	Used in
S4	IndiaLaw: Consumer Protection (E-Commerce) Rules, 2020	Rules of 2020	"The officer must acknowledge complaints within 48 hours and resolve them within one month."	Summary, 1
S5	IMARC Group: Complaint Management Software Market	Base year 2025	"The global complaint management software market size reached USD 2.9 Billion in 2025. Looking forward, IMARC Group expects the market to reach USD 6.3 Billion by 2034, exhibiting a growth rate (CAGR) of 8.57% during 2026-2034."	Summary, 2
S6	360iResearch via GII: Complaint Management Software Market	Base year 2025	"Base Year [2025] USD 2.63 billion Estimated Year [2026] USD 2.89 billion Forecast Year [2032] USD 5.29 billion CAGR (%) 10.49%"	Summary, 2, 4
S7	IMARC Group: AI Complaint Management Market	Base year 2025	"The global AI complaint management market size was valued at USD 6,406.7 Million in 2025 ... expected to reach USD 33,192.6 Million by 2034, exhibiting a growth rate (CAGR) of 19.45% during 2026-2034."	2
S8	360iResearch via GII: Artificial Intelligence in Call Centers Market	Base year 2025	"Base Year [2025] USD 1.70 billion Estimated Year [2026] USD 1.98 billion Forecast Year [2032] USD 5.20 billion CAGR (%) 17.31%"	2

ID	Source	Date	Quote from the page	Used in
S9	Outlook Business (PTI): Q-Comm orders double to ₹64,000 Cr in FY25	10 Jul 2025	"Indians ordered ₹64,000 crore of goods from quick-commerce platforms like Blinkit and Instamart in FY25, more than double from the previous fiscal year's ₹30,000 crore"; "estimated to grow more than three-times to touch ₹2 lakh crore by FY28, the report by one of the arms of domestic rating agency, Careedge Ratings, said."	Summary, 3
S10	Storyboard18: India's qcomm market surges to $7B in 2024 (Flipkart and Bain & Company)	2024 data	"its total gross merchandise value (GMV) skyrocketing from $1.6 billion in 2022 to $7 billion in 2024. This rapid expansion is expected to continue at over 40% annually until 2030"	3
S11	GAEE India: Indian Quick Commerce Industry Report	May 2026	"Order Volume (Jan '26) 7.8 Mn per day ~95% YoY growth Source: RedSeer"; market-share chart: Blinkit 52%, Instamart 25%, Zepto 21%, Others 2%. Its table gives ranges: Blinkit 45–52%, Zepto 21–29%, Swiggy Instamart 25–27%. The report says it was "prepared solely for academic and research purposes".	Summary, 3

ID	Source	Date	Quote from the page	Used in
S12	eesel: Sprinklr pricing	2026	"it has since retired self-serve sign-up entirely. Today every suite is sold through a custom Enterprise quote"; "Before the change, its Advanced plan started around $249 per month"	4, 6, 10
S13	Chattermill: Best customer experience intelligence software	2026	"The entry Support Team plan starts at $19 per agent per month (billed annually), while the fuller Suite plans ... start at $55 per agent per month."; "Most enterprise-grade platforms, including Chattermill, Medallia, and Enterpret, use custom pricing based on feedback volume and requirements."; "Qualtrics does not publish enterprise CX pricing"; "Unwrap AI clusters feedback by semantic meaning"	4, 6, 10
S14	OnClarity: Qualtrics alternatives	Updated 28 Aug 2026	"a per-agent add-on charged on top of the underlying Zendesk Suite subscription"	4, 10

Our own evidence (public repository Kartikkatwe/Hackstellar):

ID	File	What it holds
R1	backend/runs/grocery_report.json	Live run of 13 Nov 2026: 1,500 complaints, real Jev and Claude Haiku 5.5. Funnel shares, cost per step, accuracy against the answer key, discovered issues

ID	File	What it holds
R2	HANDOFF.md §4	Latencies (0.55 s median), cost by monthly volume, baseline calibration on 12 real calls, detection times
R3	SCALABILITY.md	Scalability design, bottlenecks and phases
R4	HOW_IT_WORKS.md	Every file and function, the business-logic numbers, weak spots (§12)

Checked and excluded. These figures appeared in search results, but could not be verified on the source page, so they are not used:

Claim	Why excluded
Helpline averaged 1,13,551 dockets a month (Apr–Dec 2024)	Not in the PIB document; the original parliamentary source was not opened
Customer experience management market $14.0–22.4B (2025)	MarketsandMarkets and Fortune Business Insights refused automated access; KBV Research did not respond
Complaint management software $2.84B (Coherent) and $5.2B (Orion)	Coherent's page refused access; Orion's was not opened, and its figure is about double the others
$13.50 median cost of an assisted contact (Gartner)	Gartner's page refused automated access
Medallia median contract about $13,338 a year	Not found on the cited page
Quick commerce at 2.5M orders a day	Secondary aggregator, not opened
RBI Ombudsman annual report itself	Page unreachable; its figures are cited through S3